Ex.No. 14: 5G Throughput vs SNR with Modulation Labels Date

AIM:

To simulate the performance of a 5G wireless communication system using MATLAB by analyzing the relationship between Throughput (Mbps), Signal-to-Noise Ratio (SNR) (dB), and different Modulation Schemes (QPSK, 16-QAM, 64-QAM, and 256-QAM).

Objective:

1. To evaluate the Throughput (Mbps) of a 5G communication system under different Signal-to-Noise Ratio (SNR) conditions.

2. To analyze the effect of Signal-to-Noise Ratio (SNR) (dB) on the performance and reliability of the 5G communication system.

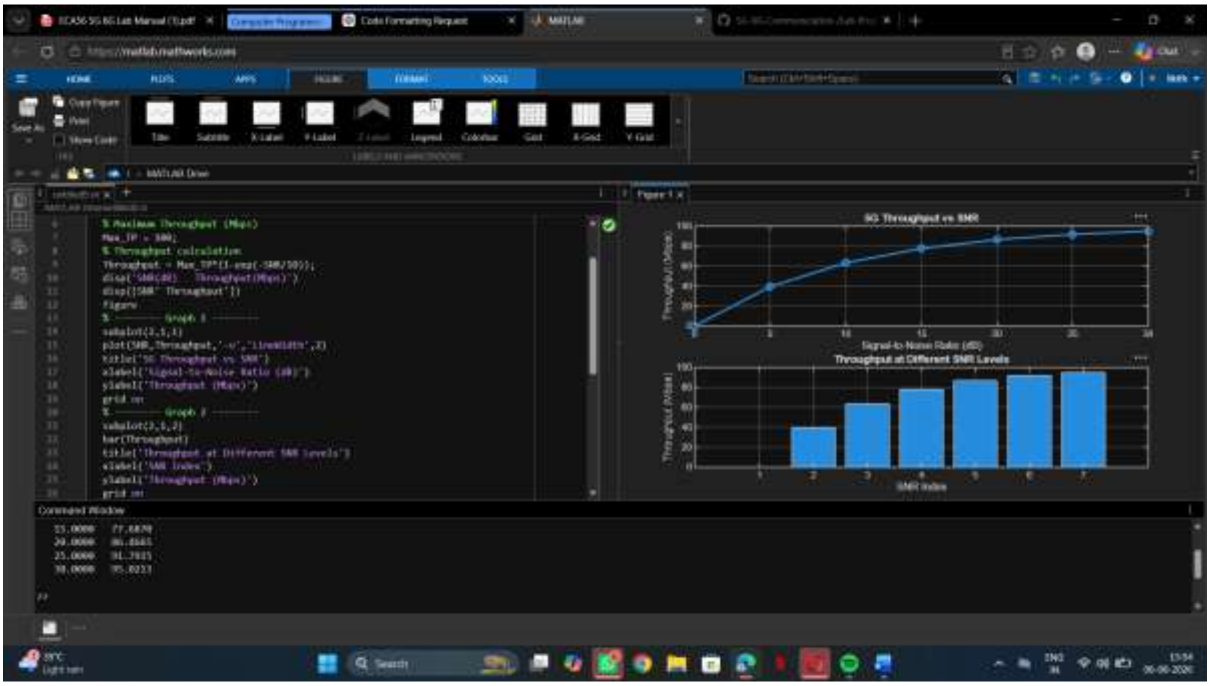
3. To compare the performance of different Modulation Schemes (QPSK, 16-QAM, 64 QAM, and 256-QAM) and determine their suitability for varying SNR conditions.

Objective 1 Matlab Code:

```
clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Maximum Throughput (Mbps)

Max_TP = 100;

% Throughput calculation

Throughput = Max_TP * (1 - exp(-SNR/10));

disp('SNR(dB)   Throughput(Mbps)')

disp([SNR' Throughput'])

figure;

% -------- Graph 1 --------

subplot(2,1,1)

plot(SNR, Throughput, '-o', 'LineWidth', 2)

title('5G Throughput vs SNR')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Throughput (Mbps)')

grid on

% -------- Graph 2 --------
```

```
subplot(2,1,2)

bar(Throughput)

title('Throughput at Different SNR Levels')

xlabel('SNR Index')

ylabel('Throughput (Mbps)')

grid on
```

OUTPUT :

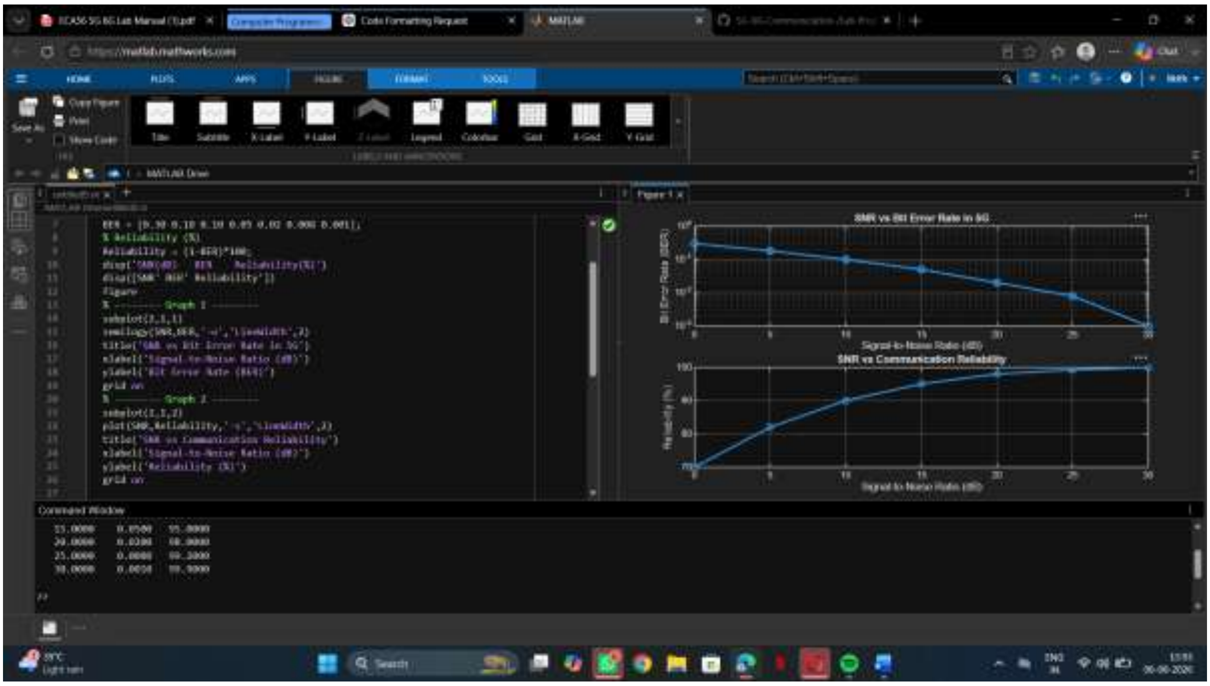


Objective 2 Matlab Code:

```
clc;

clear;

close all;

% SNR values (dB)

SNR = 0:5:30;

% Simulated Bit Error Rate (BER)

BER = [0.30 0.18 0.10 0.05 0.02 0.008 0.001];

% Reliability (%)

Reliability = (1 - BER) * 100;

disp('SNR(dB)   BER    Reliability(%)')

disp([SNR' BER' Reliability'])
```

```
figure;

% -------- Graph 1 --------

subplot(2,1,1)

semilogy(SNR, BER, '-o', 'LineWidth', 2)

title('SNR vs Bit Error Rate in 5G')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Bit Error Rate (BER)')

grid on

% -------- Graph 2 --------

subplot(2,1,2)

plot(SNR, Reliability, '-s', 'LineWidth', 2)

title('SNR vs Communication Reliability')

xlabel('Signal-to-Noise Ratio (dB)')

ylabel('Reliability (%)')

grid on
```

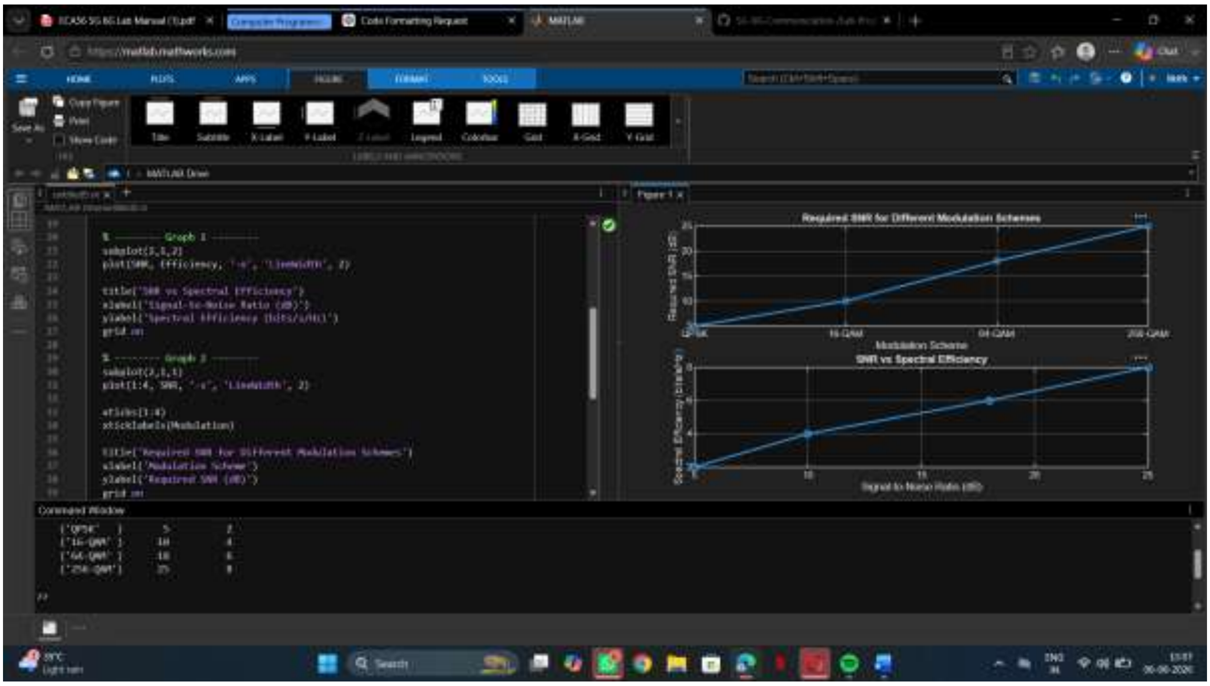
OUTPUT :



Objective 3 Matlab Code:

```
clc;
clear;
close all;
% Modulation schemes
Modulation = {'QPSK','16-QAM','64-QAM','256-QAM'};
% Required SNR (dB)
SNR = [5 10 18 25];
% Spectral Efficiency (bits/s/Hz)
Efficiency = [2 4 6 8];
disp('Required SNR (dB) and Spectral Efficiency')
disp(table(Modulation', SNR', Efficiency', ...
   'VariableNames', {'Modulation','SNR_dB','Efficiency'}))
figure;
% -------- Graph 1 --------
subplot(2,1,2)
plot(SNR, Efficiency, '-o', 'LineWidth', 2)
title('SNR vs Spectral Efficiency')
xlabel('Signal-to-Noise Ratio (dB)')
ylabel('Spectral Efficiency (bits/s/Hz)')
grid on
% -------- Graph 2 --------
subplot(2,1,1)
plot(1:4, SNR, '-s', 'LineWidth', 2)
xticks(1:4)
xticklabels(Modulation)
title('Required SNR for Different Modulation Schemes')
xlabel('Modulation Scheme')
ylabel('Required SNR (dB)')
grid on
```

OUTPUT :



RESULT :

1. The Throughput (Mbps) of the 5G communication system increased with increasing Signal-to-Noise Ratio (SNR), demonstrating improved data transmission performance.
2. The simulation showed that higher SNR (dB) reduced transmission errors and enhanced the reliability and quality of 5G communication.
3. The comparison of QPSK, 16-QAM, 64-QAM, and 256-QAM revealed that higher order modulation schemes achieved greater throughput and spectral efficiency but required higher SNR for reliable communication.